# MONOMORPHISMS AND EPIMORPHISMS VIA THE SELF FIBRE PRODUCT AND PUSHOUT

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*Editorial note.* Counted span is the author's own proof environment, source lines 1449-1481, following the statement at 1437-1447.

*Editorial note.* The dual second part (lines 1479-1480, phrase: the proof is exactly the same for the second point) is retained verbatim as a mirror-level omission; the executed part (1) covers its dual.

*Editorial note.* The converse direction assumes the isomorphism X x_Y X = X with wording retained verbatim from the source.

*Allowed standard prerequisites (editorial).*

- Universal property of the fibre product. the commutative square with sides id_X displayed at proof lines 1454-1459 and 1466-1469. Supplies the criterion for X to be the fibre product X x_Y X used in both directions of part (1).
- Definitions of monomorphism and epimorphism. the cancellation conditions on alpha, beta used at proof lines 1450-1452 and 1470-1476. Supplies the two directions: mono gives alpha = beta, and the unique gamma forces alpha = beta.

**Lemma 0.1.** *Let $\mathcal{C}$ be a category, and let $f : X \to Y$ be a morphism of $\mathcal{C}$. Then*

(1) *$f$ is a monomorphism if and only if $X$ is the fibre product $X \times_Y X$, and*
(2) *$f$ is an epimorphism if and only if $Y$ is the pushout $Y \amalg_X Y$.*

*Proof.* Suppose that $f$ is a monomorphism. Let $W$ be an object of $\mathcal{C}$ and $\alpha, \beta \in \mathrm{Mor}_{\mathcal{C}}(W, X)$ such that $f \circ \alpha = f \circ \beta$. Therefore $\alpha = \beta$ as $f$ is monic. In addition, we have the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\mathrm{id}_X} & X \\ {\scriptstyle \mathrm{id}_X}\downarrow & & \downarrow{\scriptstyle f} \\ X & \xrightarrow{f} & Y \end{array}$$

which verify the universal property with $\gamma := \alpha = \beta$. Thus $X$ is indeed the fibre product $X \times_Y X$.

Suppose that $X \times_Y X \cong X$. The diagram

$$\begin{array}{ccc} X & \xrightarrow{\mathrm{id}_X} & X \\ {\scriptstyle \mathrm{id}_X}\downarrow & & \downarrow{\scriptstyle f} \\ X & \xrightarrow{f} & Y \end{array}$$

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commutes and if $W \in \mathrm{Ob}(\mathcal{C})$ and $\alpha, \beta : W \to X$ such that $f \circ \alpha = f \circ \beta$, we have a unique $\gamma$ verifying

$$\gamma = \mathrm{id}_X \circ \gamma = \alpha = \beta$$

which proves that $\alpha = \beta$.

The proof is exactly the same for the second point, but with the pushout $Y \amalg_X Y = Y$. $\square$